

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Machine Learning [CSE-4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast classification and regression tasks in machine learning. Provide examples of each type of task and discuss the specific objectives and evaluation metrics associated with them (5)
 - A)
 - B) Evaluate the effectiveness of ensemble methods such as bagging and boosting in improving the robustness and accuracy of machine learning models. Discuss how these methods combine multiple base learners to overcome individual weaknesses. (5)
- 2) Contrast the mathematical principles behind PCA and ICA and explain how they lead to distinct methods for dimensionality reduction and signal separation? What advantages and limitations do these mathematical approaches offer in real-world scenarios. (5)
 - A)
 - B) Explore the concept of kernel trick and its significance in SVM. How does the kernel trick enable SVM to efficiently handle high-dimensional feature spaces without explicitly computing the transformation? Can you provide insights into the mathematical intuition behind the kernel trick. (5)
- 3) Assess the limitations of K-means clustering and propose alternative clustering algorithms that address these limitations. Evaluate the performance of the proposed algorithms in terms of scalability, robustness, and ability to handle non-linear data distributions. (5)
 - A)
 - B) Analyze the relationship between the bias-variance tradeoff and model complexity. How does increasing model complexity affect the bias and variance components of the error, and how does this impact the overall model performance. (5)
- 4) Analyze the challenges of applying reinforcement learning algorithms in real-world scenarios with large state and action spaces. Discuss strategies for dealing with the curse of dimensionality and improving the efficiency of value function approximation and policy search methods. (5)
 - A)
 - B) Compare and Contrast value iteration and policy iteration algorithms. How does the choice of discount factor affect the convergence and stability of these algorithms. (5)
- 5) Evaluate the impact of data imbalance on the performance of classification algorithms such as logistic regression and support vector machines. Discuss techniques for handling (5)

imbalanced datasets and their implications for model fairness and interpretability.

A)

B)

Consider the tuple $X = (\text{Weather} = \text{Rainy}, \text{Temperature} = \text{Mild}, \text{Humidity} = \text{High}, \text{Windy} = \text{True})$ Calculate the conditional probabilities required for predicting the class label 'Outcome' for both 'Win' and 'Loss.'

Weather	Temperature	Humidity	Windy	Outcome
Sunny	Hot	High	False	Win
Sunny	Hot	High	True	Win
Overcast	Hot	High	False	Win
Rainy	Mild	High	False	Win
Rainy	Cool	Normal	False	Lose
Rainy	Cool	Normal	True	Lose
Overcast	Cool	Normal	True	Win
Sunny	Mild	High	False	Win
Sunny	Cool	Normal	False	Win
Rainy	Mild	Normal	False	Lose

(5)

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